

Facility and Infrastructure Optimization for a Renewable-Energy Company

Exact Mixed-Integer Programming versus Metaheuristics on Three Operations-Research Problems

Syed Abbas Ahmad

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Abstract

We study three classic operations-research problems arising from a single renewable-energy company dataset: (P1) laying out 30 departments across 197 candidate sites as a Quadratic Assignment Problem; (P2) siting electrical substations across 137 demand zones as a Capacitated Facility Location Problem; and (P3) assigning 20 products to 25,189 storage slots to minimize material-handling travel. Each problem is solved both *exactly* (mixed-integer or linear programming with the CBC solver via PuLP) and with *metaheuristics* (genetic algorithm, simulated annealing, particle-swarm optimization), and the approaches are compared on solution quality, runtime, and optimality gap. The exact solvers provide provably-optimal baselines where tractable; the metaheuristics recover near-optimal solutions at scale, with simulated annealing and the genetic algorithm reaching within 1.21% of optimality on the substation problem while particle-swarm trails at 16.67%. On the full department-layout QAP, where exact optimization is intractable, a Gilmore–Lawler lower bound together with a random-restart analysis brackets the metaheuristics’ quality. For storage, the Cube-per-Order-Index rule is shown to be provably optimal and to save 7.4% of travel over a realistic popularity-based policy.

1 Introduction

The dataset (`Group 1.xlsx`) describes a renewable-energy company whose facilities include turbines, photovoltaic maintenance centers, grid-connection points, substations, and battery storage. It encodes three independent decision problems that map cleanly onto canonical OR formulations. Our goals are (i) rigorous, reproducible models; (ii) provably-optimal baselines where tractable; and (iii) an honest comparison of exact and heuristic methods. We pair exact mixed-integer and linear programming (PuLP [12] with the CBC solver [4]) with three metaheuristics: a genetic algorithm [5], simulated annealing [9], and particle-swarm optimization [8]. All code, data, and logged results accompany this report; every figure and number is regenerated from `output/results/` by seeded scripts.

2 Data

The data comprises a 30×30 asymmetric department flow matrix; 197 candidate locations with coordinates and a type label; 137 demand zones with electricity demand, construction cost, capacity, and a 137×137 (asymmetric) distance matrix with forbidden links; and 20 products with throughput and storage requirements. Parsing, validation, and export are handled by `data/loader.py`.

Modeling assumptions, including a unit inconsistency in the substation data that renders the literal reading infeasible, are documented in `docs/formulations.md` (assumptions A1–A8) and summarized in Section 7.

3 P1 – Department Layout (QAP)

3.1 Model

Assign departments $\mathcal{D}$ to candidate locations $\mathcal{L}$ to minimize total weighted flow $\times$ distance — the Koopmans–Beckmann quadratic assignment problem [10, 11],

$$\min \sum_{i,j \in \mathcal{D}} \sum_{k,\ell \in \mathcal{L}} f_{ij} d_{k\ell} x_{ik} x_{j\ell}, \quad \sum_k x_{ik} = 1 \ \forall i, \quad \sum_i x_{ik} \leq 1 \ \forall k,$$

with optional type-compatibility $x_{ik} \leq a_{ik}$. Distances are Euclidean from the site coordinates. Because $|\mathcal{L}| = 197 > |\mathcal{D}| = 30$, this is a QAP with location selection.

3.2 Methods and results

The full instance is intractable exactly ($O(|\mathcal{D}|^2|\mathcal{L}|^2)$ linearization terms), so we obtain the optimum of a reduced 8×8 instance by full enumeration (2,874,111); both the GA and SA recover it exactly. Notably, a generic McCormick-linearized MILP solved with CBC *stalls* on the same instance, returning 3,032,631 ($\approx 5.52\%$ above the true optimum) while reporting status “optimal” — a concrete reminder that the QAP defeats general-purpose branch-and-cut even at small size [2]. On the full $30 \rightarrow 197$ problem the metaheuristics run alone: simulated annealing reaches 4,928,976 and the genetic algorithm 5,038,900, SA outperforming GA as expected for QAP-type problems.

Because no exact optimum is available at this scale, we characterize solution quality three complementary ways. A Gilmore–Lawler lower bound [11] certifies that the optimum is at least 3,199,175, placing SA within 54.1% of optimal; the GL bound is notoriously loose for the QAP, so this is a conservative certificate rather than a tight one. Two further signals indicate the layout is genuinely strong: SA improves on the mean of 5000 random feasible layouts by 80.2%, and its seed-to-seed standard deviation is only 1.9% — evidence of a stable, near-optimal basin rather than a lucky single run. Figure 1 shows the resulting layout and convergence (with the GL bound and random reference marked).

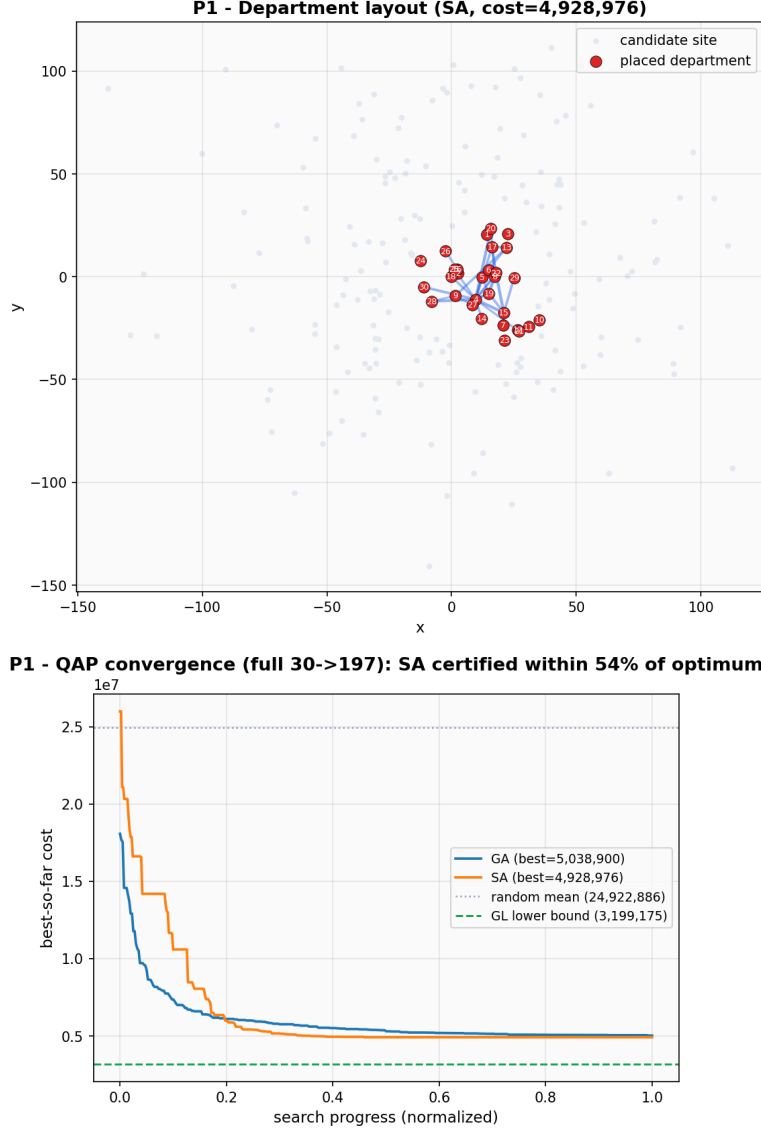


Figure 1: P1: best department layout (top) with high-flow pairs drawn as ribbons, and GA-vs-SA convergence on the full instance (bottom).

4 P2 – Substation Siting (CFLP)

4.1 Model

Open substations and assign zones to minimize construction plus transmission cost — a capacitated facility-location problem [1, 3],

$$\min \sum_s c_s y_s + \alpha \sum_{z,s} d_{zs} q_z x_{zs}, \quad \sum_s x_{zs} = 1, \quad x_{zs} \leq y_s, \quad \sum_z q_z x_{zs} \leq \kappa_s y_s,$$

single-source, with forbidden links omitted and a self-service diagonal ($d_{ss} = 0$). The transmission rate α is calibrated from the data so that construction and transmission are comparable.

4.2 Methods and results

CBC solves the 137-zone instance to optimality (objective 877,868, 109 substations built). Relative to a do-nothing “build-everywhere” policy — a substation in every zone, each self-serving, at cost 1,033,711 — the optimized plan saves 15.1% by consolidating demand onto fewer substations. The metaheuristics, run across 10 seeds, give best objectives of 888,504 (GA), 888,504 (SA), and 1,024,250 (PSO), i.e. gaps of 1.21% (GA/SA) and 16.67% (PSO) to optimality. Particle-swarm, designed for continuous spaces, is the weakest on this binary problem even with a sigmoid decoding. An α -sweep (Figure 2) traces the classic facility-location trade-off: as the transmission rate rises, the optimal number of substations increases and transmission cost falls.



Figure 2: P2: optimal number of substations and the construction/transmission cost split as a function of the transmission rate α .

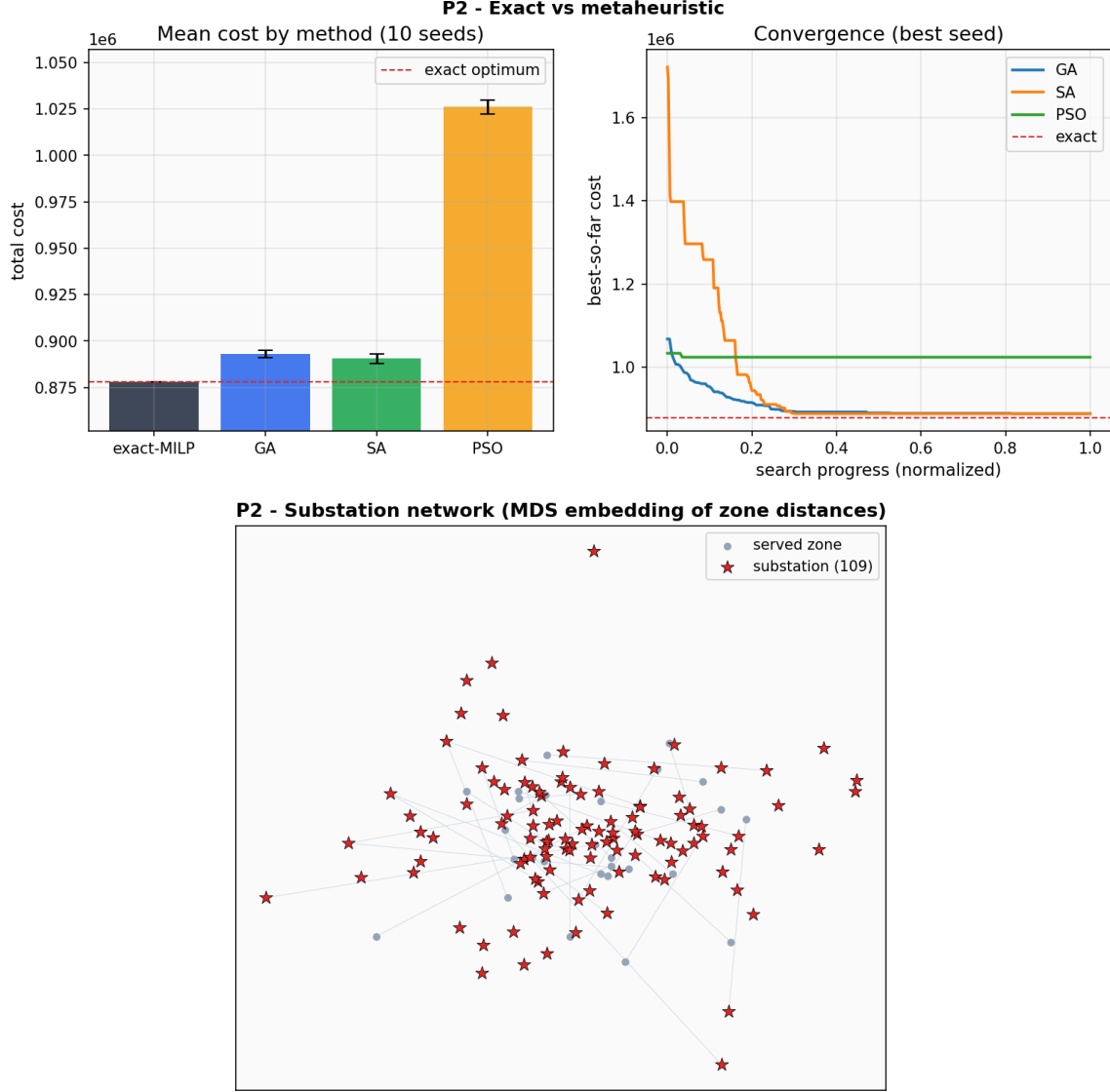


Figure 3: P2: exact-vs-metaheuristic costs and convergence (top), and the chosen substation network on an MDS embedding of the zone-distance matrix (bottom).

5 P3 – Storage Layout (Cube-per-Order Index)

5.1 Model

Assign products to unit-capacity slots to minimize expected travel,

$$\min \sum_p \sum_s \frac{t_p}{q_p} \delta_s x_{ps}, \quad \sum_s x_{ps} = q_p, \quad \sum_p x_{ps} \leq 1.$$

The constraint matrix is totally unimodular, so the LP relaxation is integral and its optimum equals the Cube-per-Order-Index (COI) ranking rule [7, 6].

5.2 Methods and results

The COI ordering yields optimal travel 40,480,413 at full scale. We verify optimality three independent ways: an exact transportation LP on a downscaled instance matches the COI value to machine precision, and both GA and SA over the product ordering reach the same optimum (0.00% gap). COI saves 27.3% of travel relative to a random layout, and — more meaningfully — 7.4% relative to a realistic popularity (throughput-only) policy (43,732,020) that ignores product volume (Figure 4).

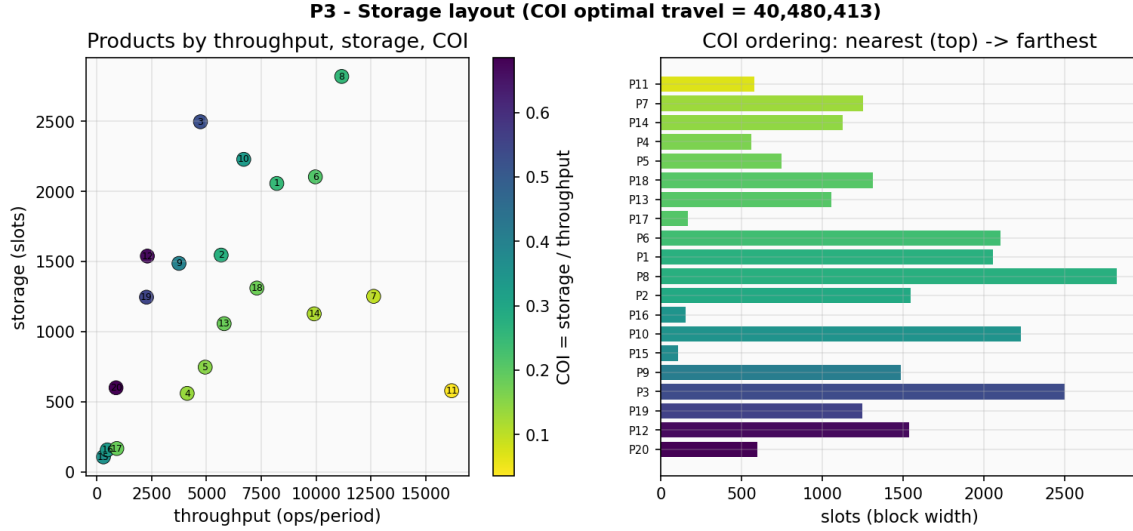


Figure 4: P3: products by throughput, storage, and COI (left), and the optimal nearest-to-farthest slot allocation (right).

6 Comparison and Discussion

Across the three problems, exact methods provide optimal or provably-optimal-on-reduction baselines, while metaheuristics deliver near-optimal solutions at full scale. Simulated annealing is the most consistent heuristic (strong on both QAP and CFLP); the genetic algorithm is competitive on CFLP and storage but weaker on the full QAP; particle-swarm, being continuous by design, is the weakest on the binary facility-location problem. For the storage problem, structure (total unimodularity / the COI theorem) makes an exact polynomial solution available, and the metaheuristics serve only as confirmation. The optimality gaps are summarized in `output/results/comparison.csv`.

7 Assumptions

All modeling judgment calls are documented as assumptions A1–A8 in `docs/formulations.md`. The most consequential is A5/A8 for P2: the literal demand/capacity units make the instance infeasible, so raw values are treated in a common unit and a calibrated transmission rate is introduced to avoid a degenerate “build-everywhere” optimum.

8 Reproducibility

Every experiment is seeded (`experiments/config.py`) and writes results to `output/results/`. Running `run_p1`, `run_p2`, `run_p3`, then `experiments.compare` and `visualization.plot_solutions` regenerates all numbers and figures in this report.

References

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